

Riemann Sums & Definite Integrals

MATH 145 – Applied Calculus

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November 4, 2024



Riemann Sums

Consider a function $f(x)$ defined on an interval $a \leq t \leq b$, and, given a whole number n , divide the interval into n subintervals of length

$$\Delta t = \frac{(b-a)}{n} \text{ with endpoints } t_0, t_1, t_2, \dots, t_n.$$

Left Riemann Sum

The **left (regular) Riemann sum** is:

$$f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t = \sum_{i=0}^{n-1} f(t_i)\Delta t.$$

Right Riemann Sum

The **right (regular) Riemann sum** is:

$$f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t = \sum_{i=1}^n f(t_i)\Delta t.$$

Definite Integral

Definite Integral

Given a continuous function $f(x)$ on an interval $a \leq t \leq b$, the **definite integral** of $f(x)$ over $[a, b]$ is the limit:

$$\int_a^b f(t) dt = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_i) \Delta t,$$

or:

$$\int_a^b f(t) dt = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(t_i) \Delta t$$

provided these limits exist. $f(x)$ is called the **integrand** and a and b are called the **limits of integration**.

Example: Definite Integral

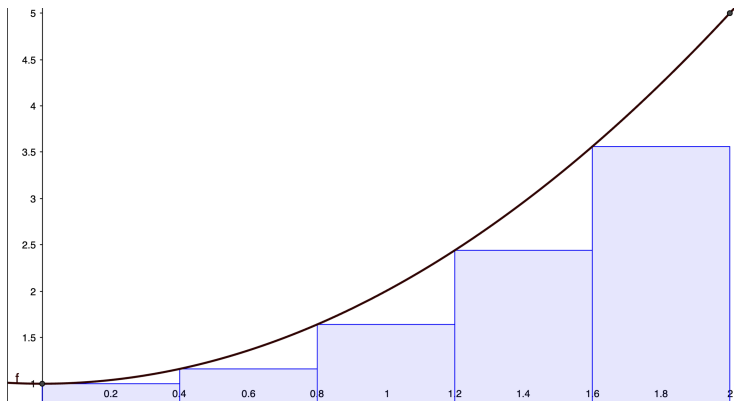
Example: Consider the function $f(x) = x^2 + 1$ over the interval $[0, 2]$. For a given n , $\Delta t = 2/n$ and $t_i = 2i/n$, so that:

$$\int_0^2 x^2 + 1 \, dx = \lim_{n \rightarrow \infty} ((2i/n)^2 + 1) \times (2/n).$$

This sum can be thought of as a sum of areas of rectangles “under” the graph of $f(x)$.

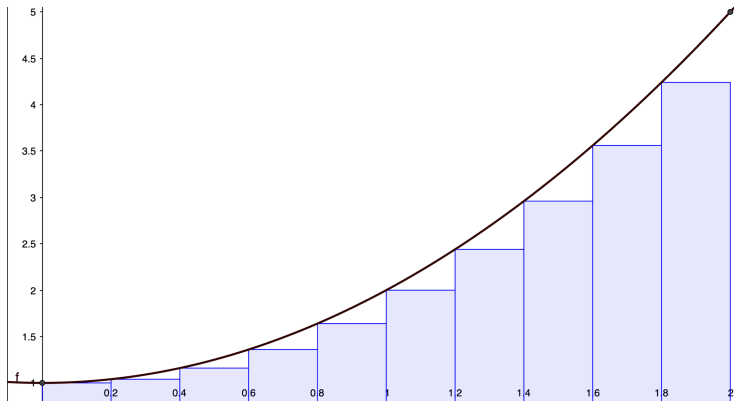
Example: $n = 5$

$$\int_0^2 x^2 + 1 \, dx = \sum_{i=0}^4 ((2i/5)^2 + 1) \times (2/5) = 4.384$$



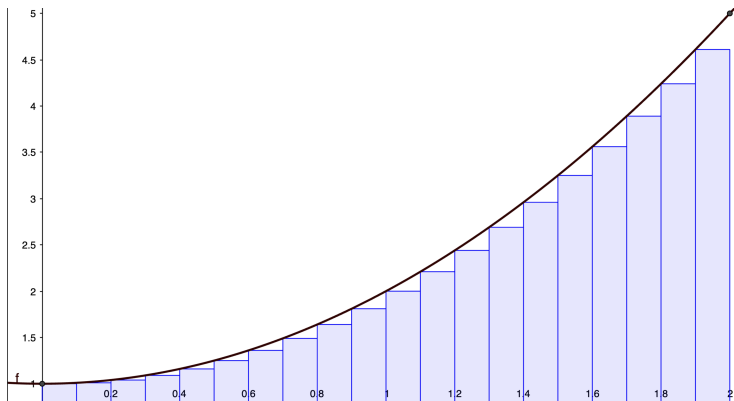
Example: $n = 10$

$$\int_0^2 x^2 + 1 \, dx = \sum_{i=0}^9 ((2i/10)^2 + 1) \times (2/10) = 4.448$$



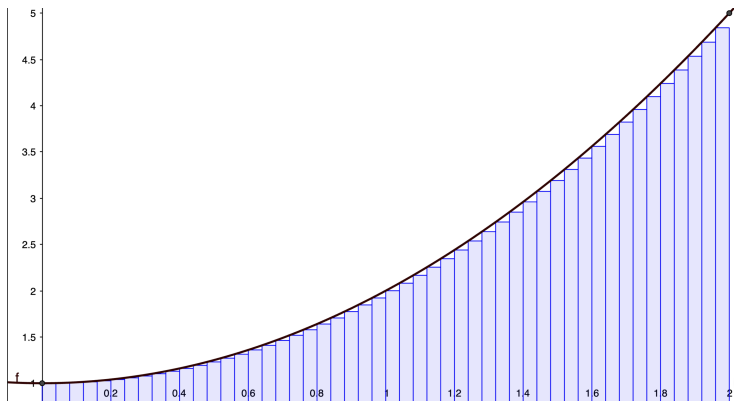
Example: $n = 20$

$$\int_0^2 x^2 + 1 \, dx = \sum_{i=0}^{19} ((2i/20)^2 + 1) \times (2/20) = 4.571$$



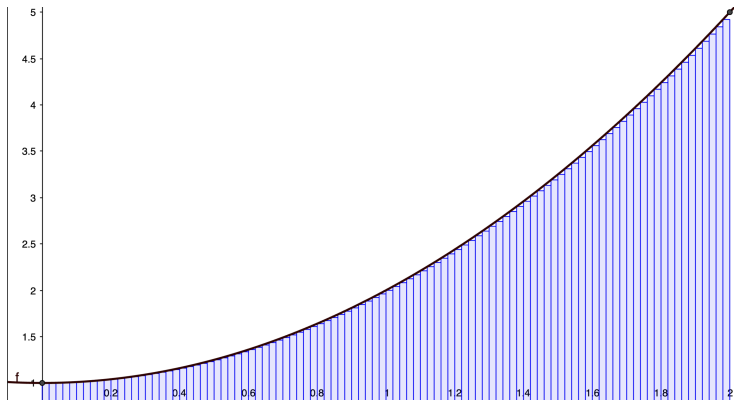
Example: $n = 50$

$$\int_0^2 x^2 + 1 \, dx = \sum_{i=0}^{49} ((2i/50)^2 + 1) \times (2/50) = 4.627264$$



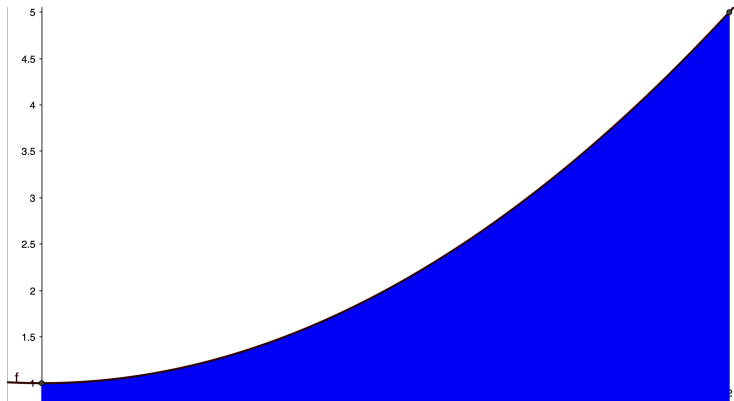
Example: $n = 100$

$$\int_0^2 x^2 + 1 \, dx = \sum_{i=0}^{99} ((2i/100)^2 + 1) \times (2/100) = 4.646808$$



Example: $n = 1000$

$$\int_0^2 x^2 + 1 \, dx = \sum_{i=0}^{999} ((2i/1000)^2 + 1) \times (2/1000) = 4.664668$$



Example: $n \rightarrow \infty$

As $n \rightarrow \infty$, this sum becomes the *area under the graph* of $f(x)$ over $[a, b]$.

$$\int_0^2 x^2 + 1 \, dx = \lim_{i \rightarrow \infty} \sum_{i=0}^{n-1} ((2i/n)^2 + 1) \times (2/n) = 14/3$$

Estimates of $\int_a^b f(x) \, dx$ can be obtained with left sums right sums, sums from midpoint values of $f(x)$ or more general types of sums.

In simple cases, the values of $\int_a^b f(x) \, dx$ can be found simply by finding an area under a graph.

Example: Definite Integral

Example: Evaluate $\int_0^2 -2x + 1 \, dx$.

$\int_a^b f(x) \, dx$ is the *signed* area – that is, the area between the graph of $f(x)$ and the x -axis.

Velocity and Distance

Recall the definition of rate of change: $r = \frac{d}{t}$.

This implies that $d = rt$.

Suppose $v(t)$ is the velocity of an object at time t and consider the distance traveled over the time interval Δt .

$$\begin{aligned}\int_a^b v(t) dt &= \lim_{n \rightarrow \infty} \sum_{i=1}^n v(t_i) \Delta t \\ &= \text{distance traveled (displacement)} \\ &= \text{area under the graph of } v(t)\end{aligned}$$

Example: Velocity and Distance

The velocity of an object moving along an axis is described by the picture below. Find the displacement of the object from time $t = 0$ to time $t = 4$.

